

Supernovae: An Explosion in Data

Matt Grayling
Institute of Astronomy

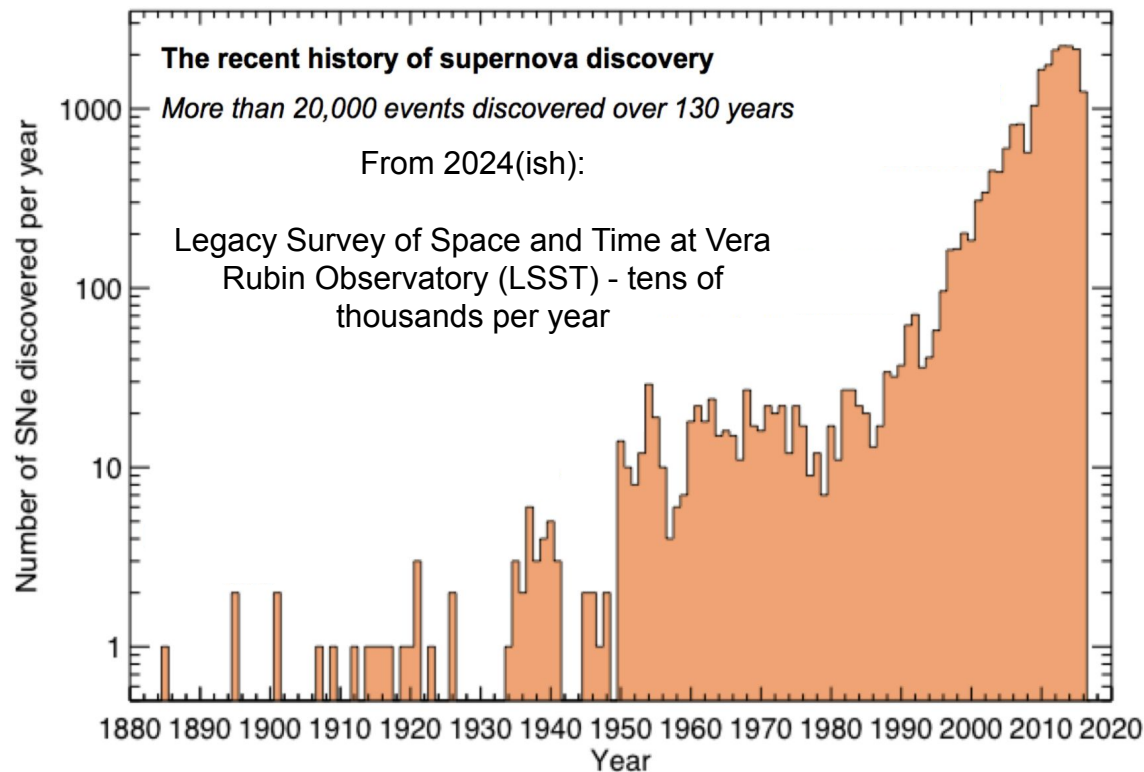


An exploration of a variety of problems in the areas of machine learning and statistical inference

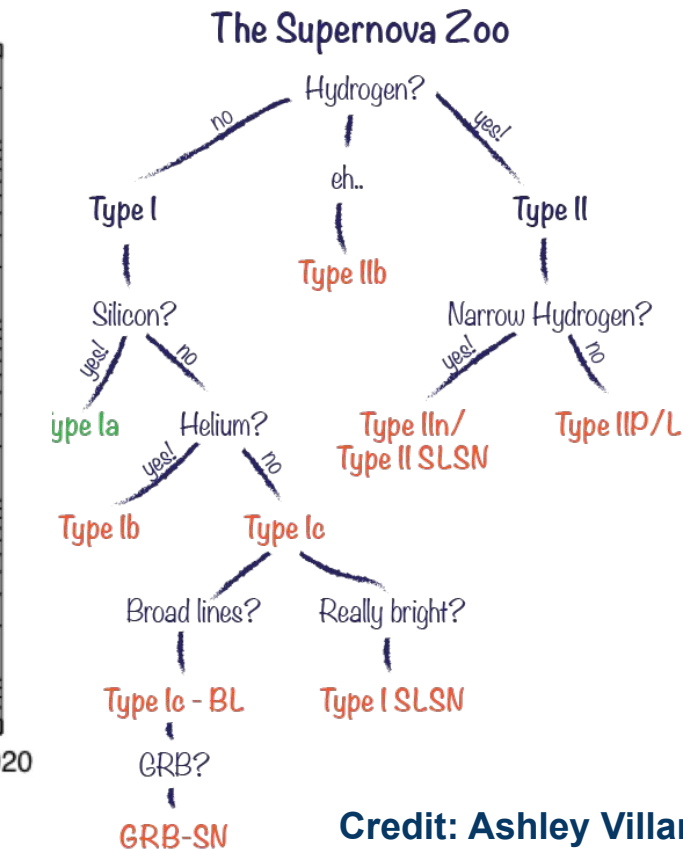
- 1) Generative models and supernova classification
- 2) Scaling Bayesian inference techniques for next generation surveys
- 3) A (brief) look at simulated-based inference

Part 1: Supernova Classification

The Problem with Supernova Classification



Credit: Mark Sullivan



Neural networks are great...

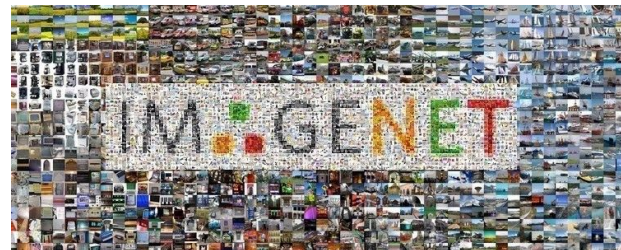
Neural network-based classification of supernova light curves has been a major research goal over the last few years e.g. PlastiCC, ElastiCC:

- Recurrent neural networks
 - RAPID (Muthukrishna+2019)
- Convolutional neural networks
 - Scone (Qu+2021)
- Bayesian neural networks
 - SuperNNova (Möller, de Boissière 2019)
- Variational autoencoders
 - ParSNIIP (Boone 2021)

Not an exhaustive list!

...if you have the data to train them

- State-of-the-art industry ML tools train on millions of examples
- For some classes of transient, we have sample sizes ~ 10 -100
- For many scientific applications, we train on large simulated data sets



ImageNet: ~ 14 million images



GPT - 4

GPT-4: 45 GB of text

- Good performance on real data requires simulations which are representative of the true population
 - Accurate simulations require good physical understanding of population
 - Supernovae are complex and diverse, simulations fail to reproduce full variability of population
 - Consistent drop in performance when applying models trained on simulations to real data

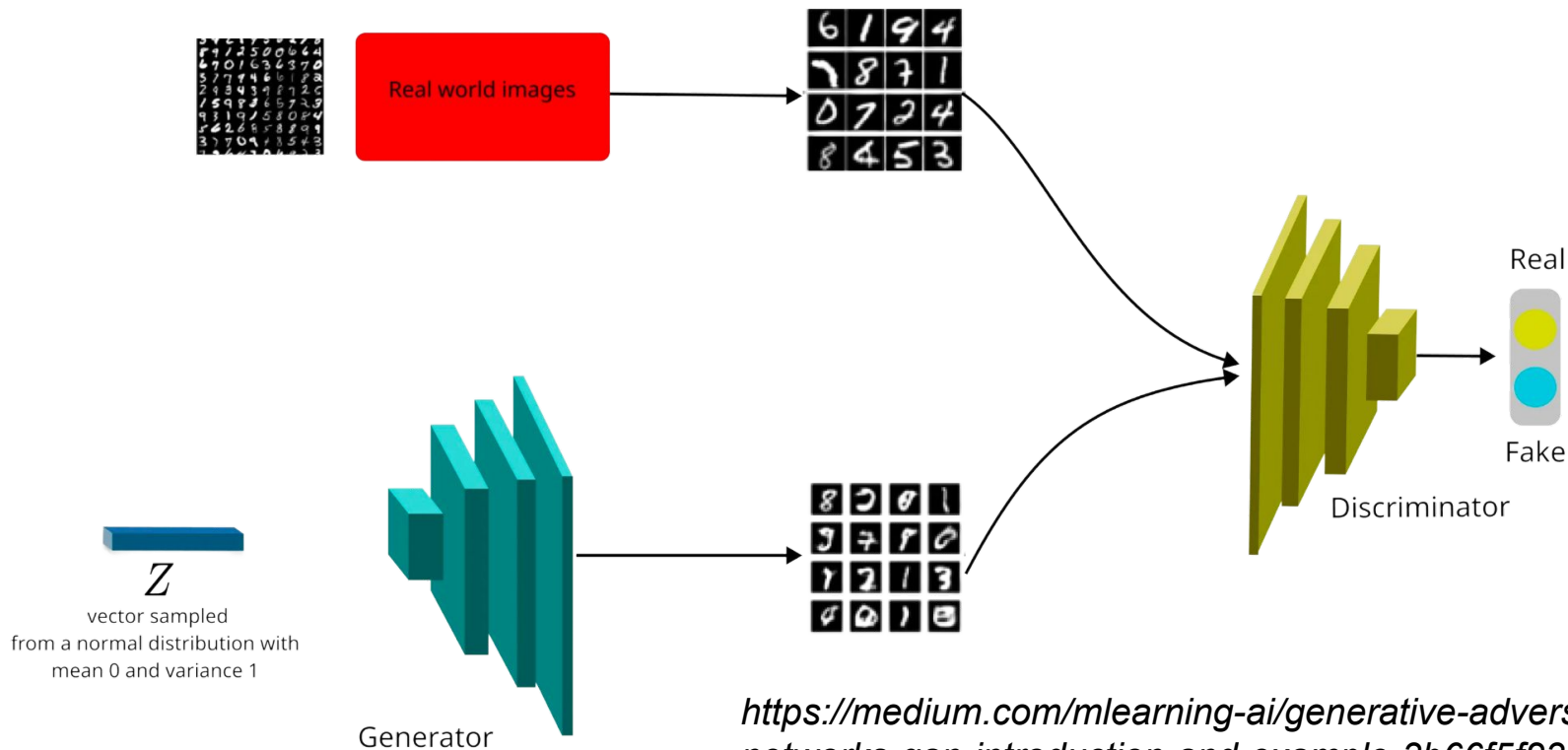
Generative Models

- Generative models allow us to sample from the distribution of supernovae, without assuming any physics
- If trained well, we can generate large synthetic data sets to train classifiers which reproduce full variability of real population
- Augmenting data-sets using generative adversarial networks (GANs) has been shown to improve classification performance within astronomy and beyond (e.g. Motamed+21, Garcia-Jara+22)



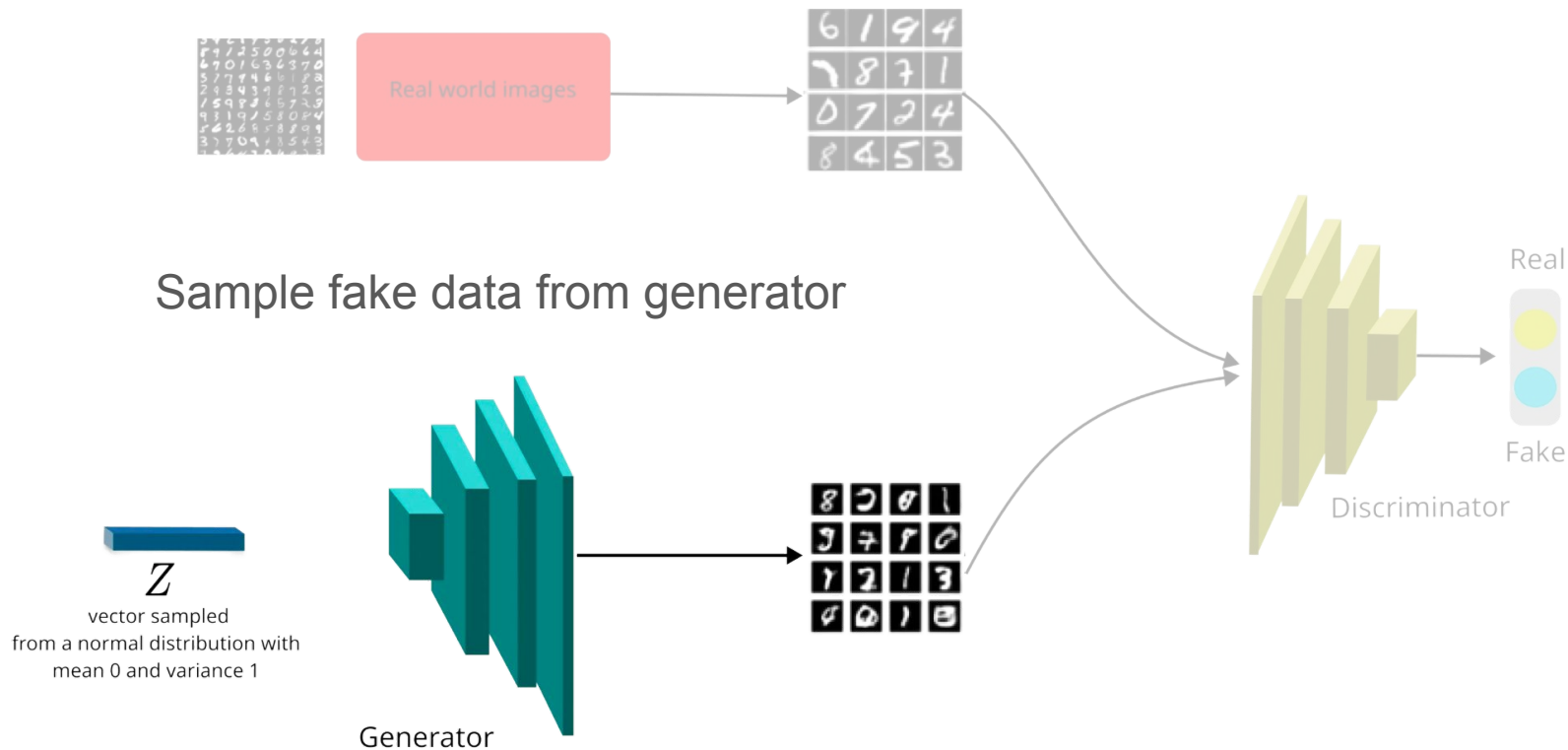
“An astronomer using machine learning to study supernovae” - Generated by DALLE 3

Generative Adversarial Networks (GANs)

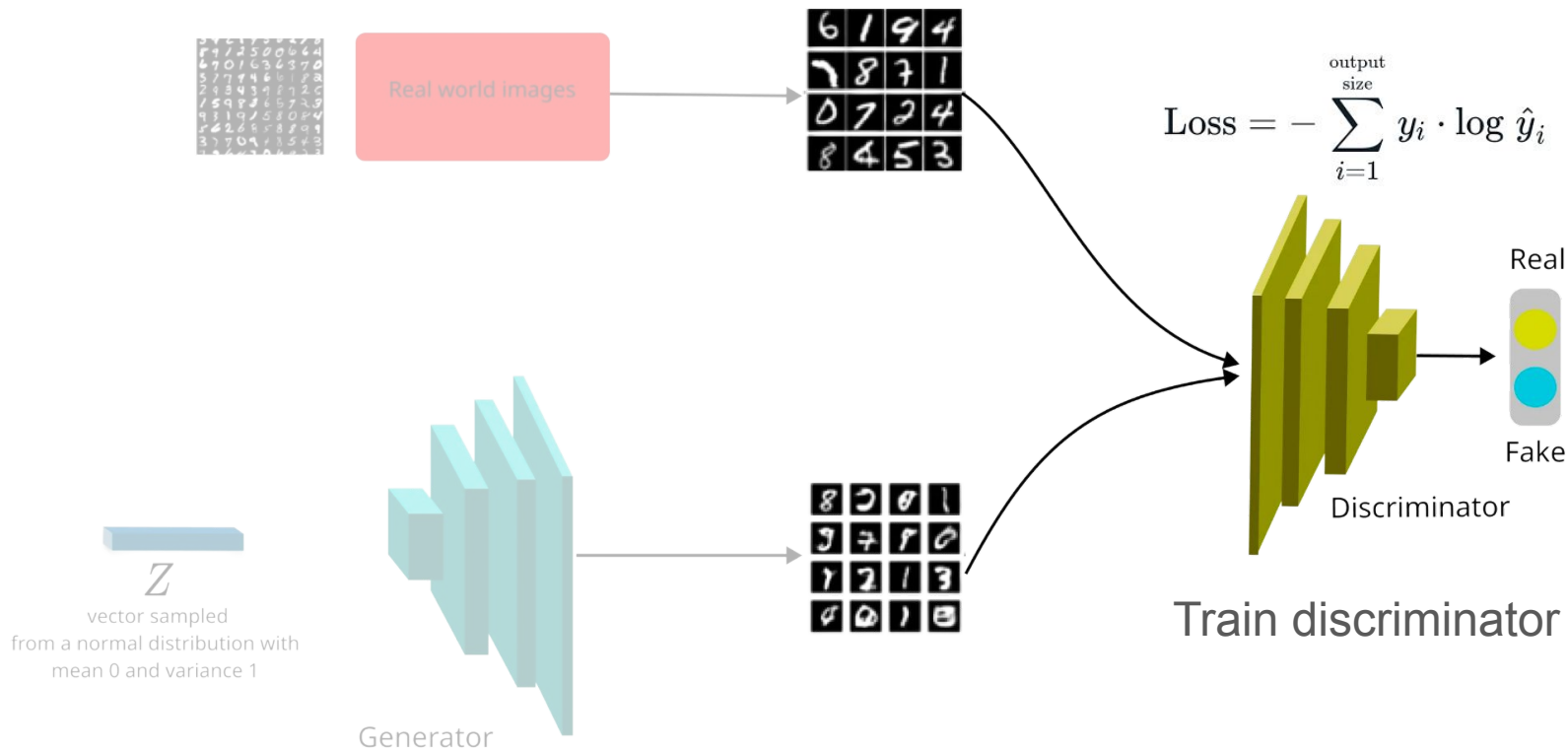


<https://medium.com/mlearning-ai/generative-adversarial-networks-gan-introduction-and-example-3b66f5f235e9>

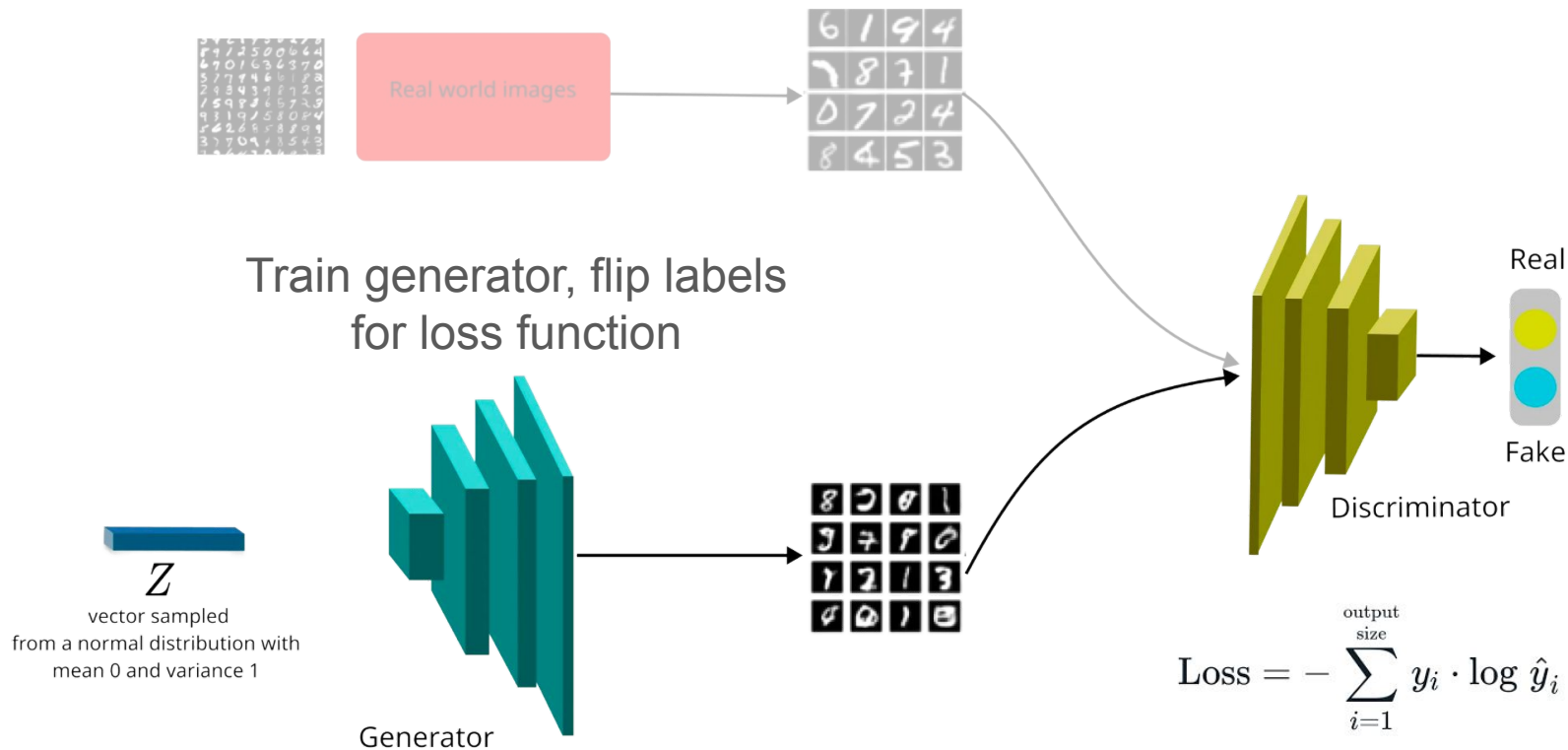
Generative Adversarial Networks (GANs)



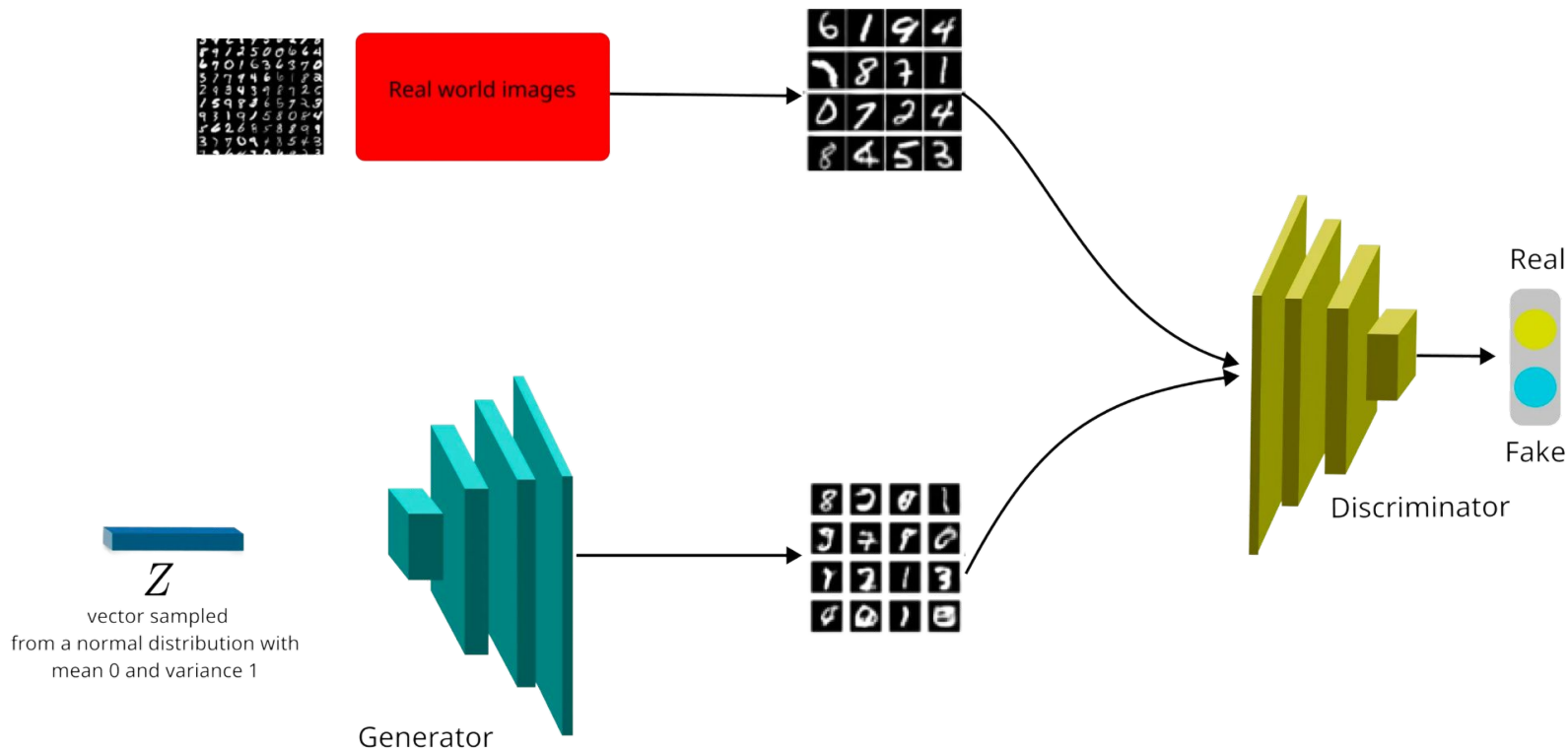
Generative Adversarial Networks (GANs)



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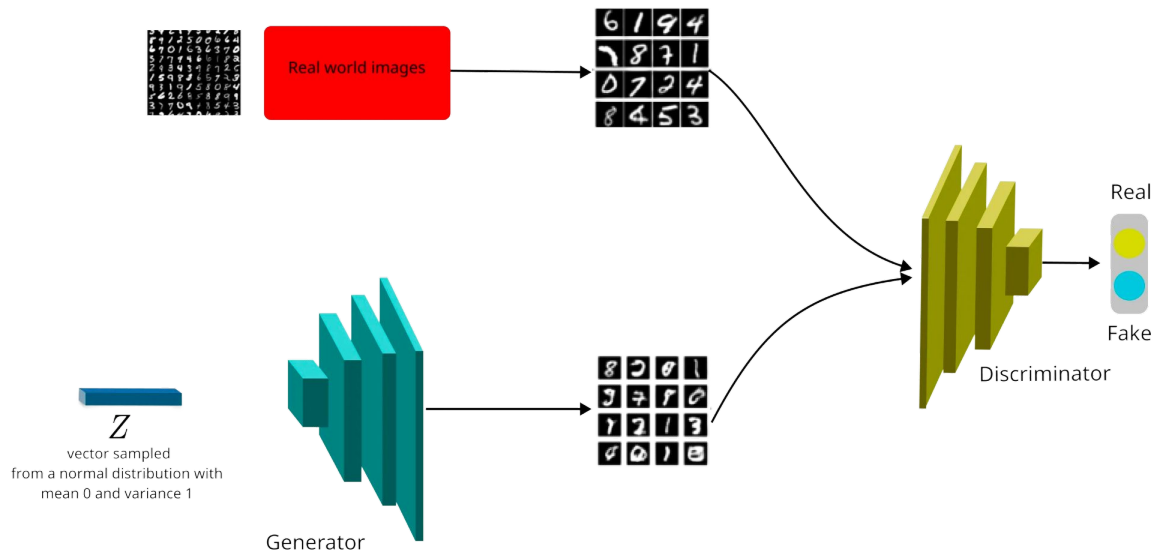


Generative Adversarial Networks (GANs)

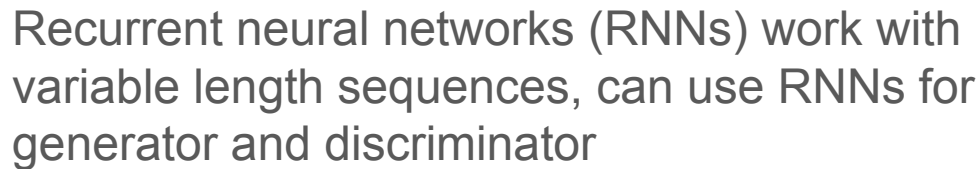


Adapting to variable length sequences

- Typically, generative models focus on generating data with a fixed structure e.g. image of fixed pixel size
- In order to generate realistic SN light curves, we need to generate variable-length time series



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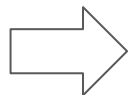
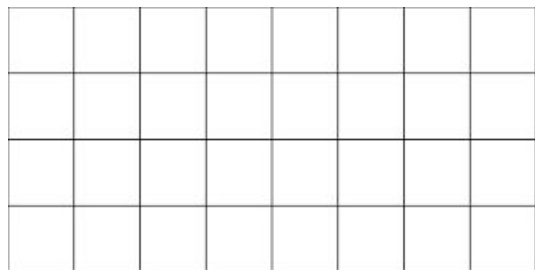


At each timestep, generate phase, photometry and uncertainty

[illegible]

Adapting to variable length sequences

Image generation, pixel grid



At each timestep, generate phase,
photometry and uncertainty

t_g	t_r	t_i	t_z	g	r	i	z	σ_g	σ_r	σ_i	σ_z
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

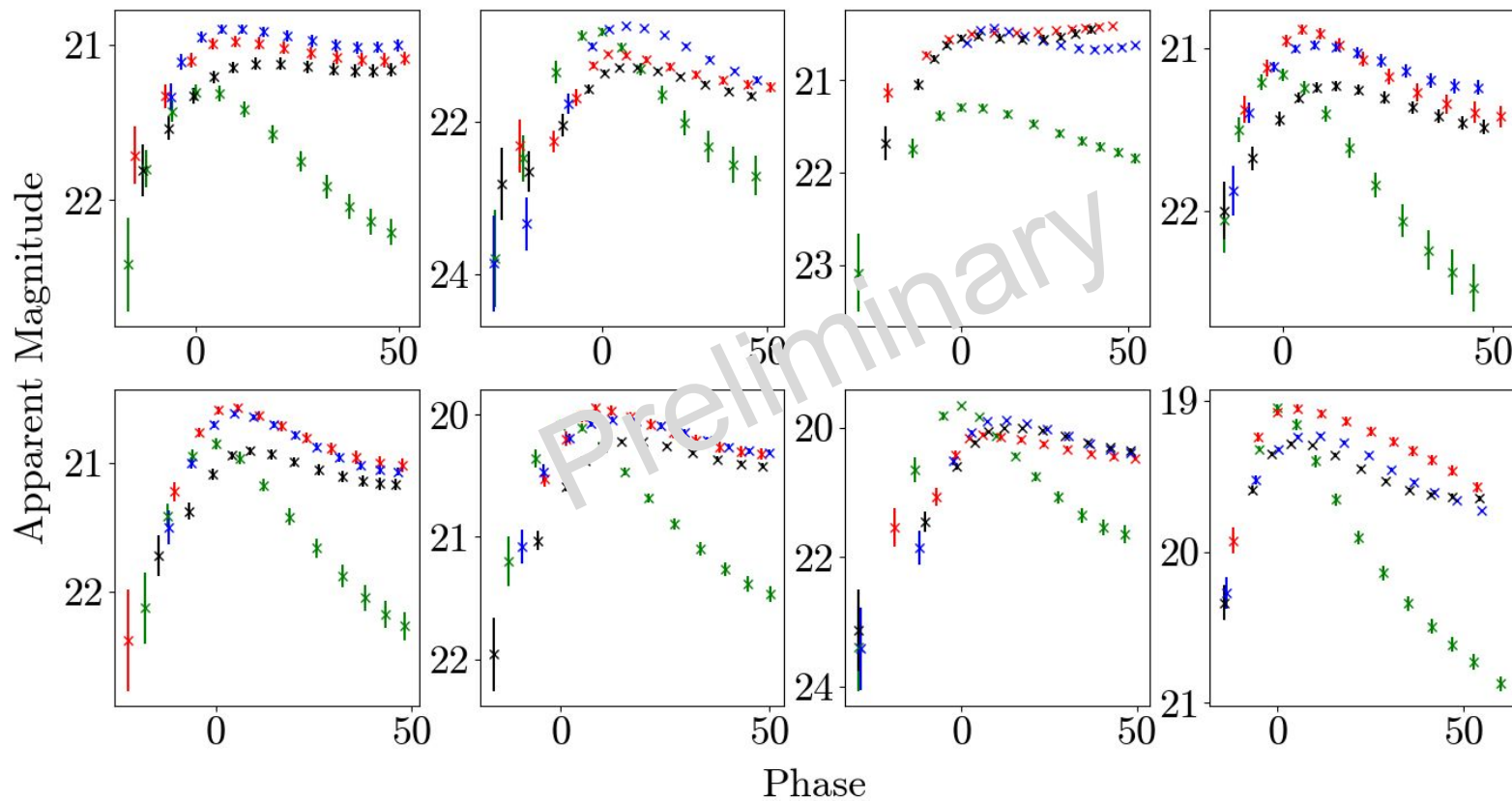
$$\mathbf{z} \sim N(\mathbf{0}, \underline{\underline{I}})$$

Input noise sampled from
n-dimensional latent space

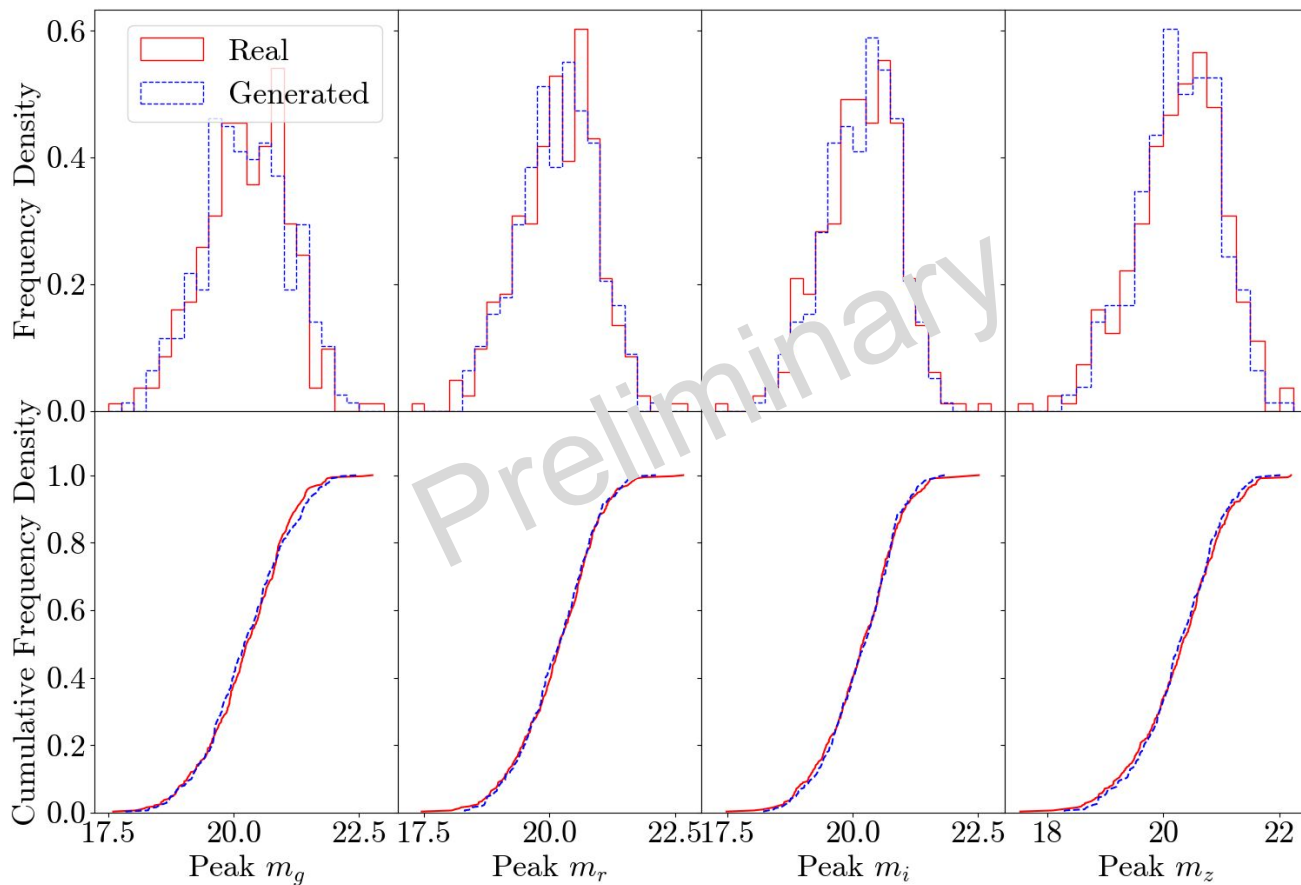
$$\underline{\underline{z}} = \begin{bmatrix} \mathbf{z} \\ \mathbf{z} \\ \mathbf{z} \\ \vdots \\ \mathbf{z} \end{bmatrix}$$

Input noise sampled from
n-dimensional latent space, then
repeated N times. Shape of input
noise determines length of time
series

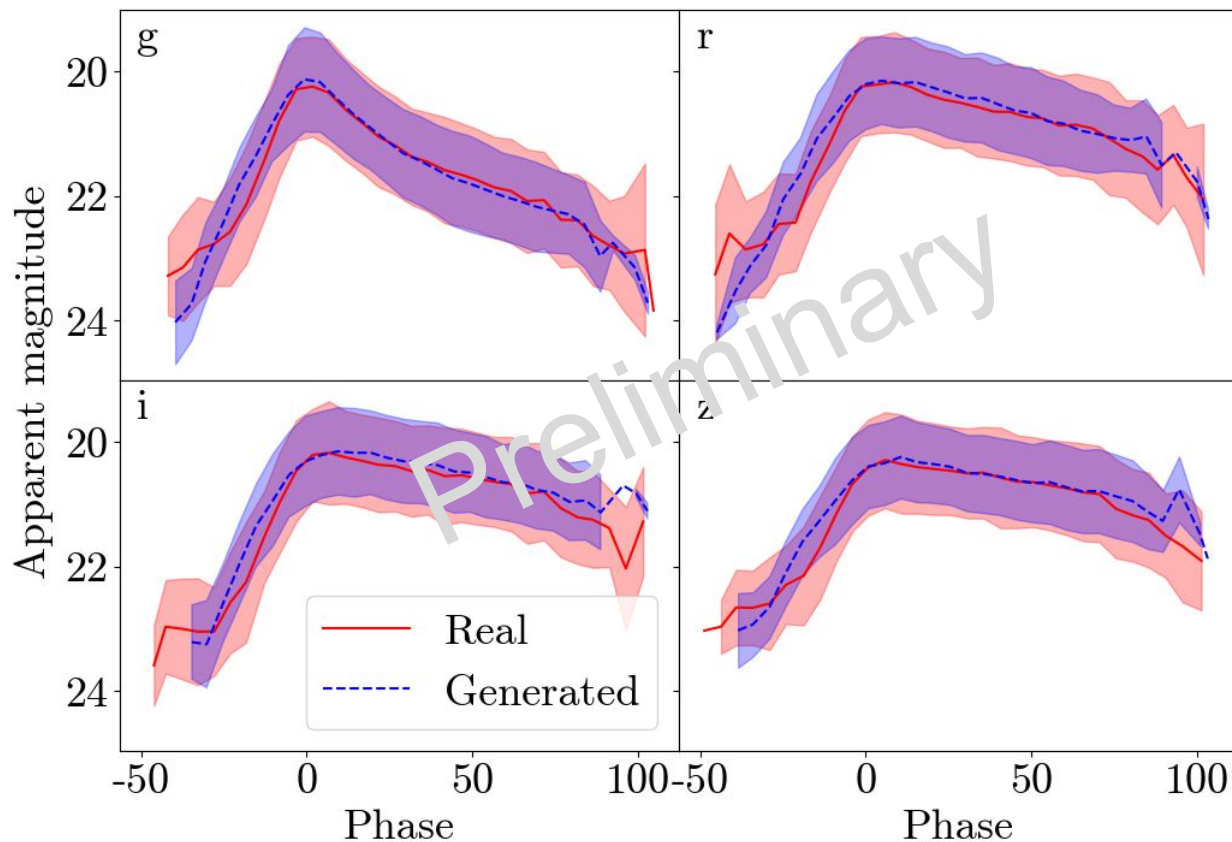
Generated Light Curve Examples



Peak Observed Brightness

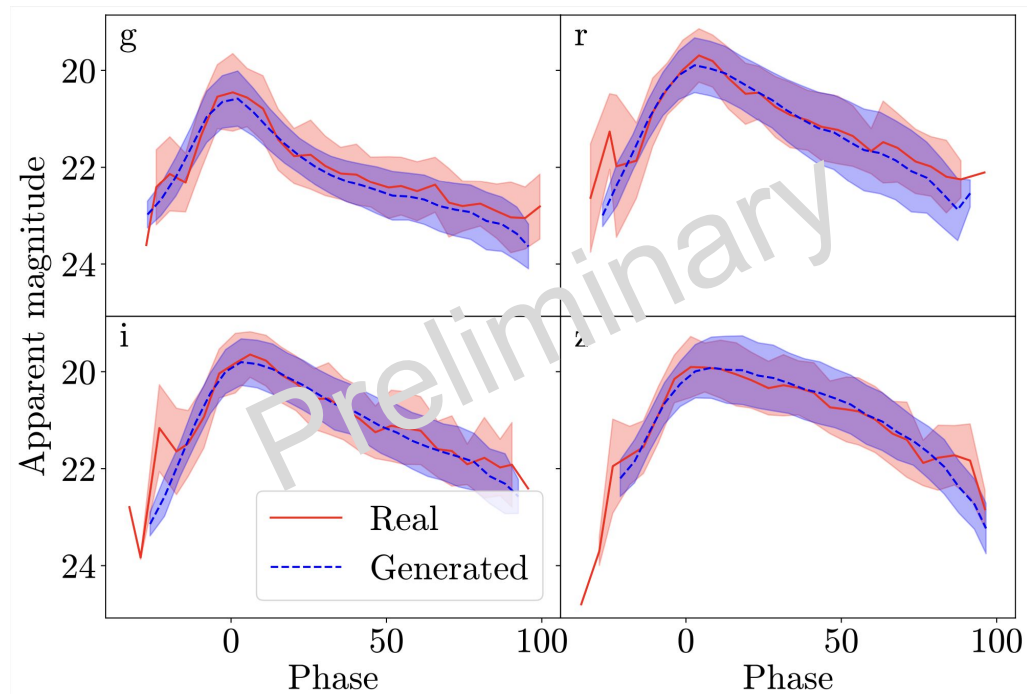


Light Curve Populations



Augmenting Rare Supernova Classes

- Previous plots were for sample of ~ 350 simulated supernovae
- Training on a much smaller data set of ~ 30 supernovae still produces comparable results
- For rare SN classes with few examples, this approach has the potential to augment very sparse training sets



- GANs have the ability to generate realistic supernova light curves and could help bridge the gap between synthetic and real data
- Next steps:
 - Train a classifier with generated data and assess performance
 - Retrain on real data
 - Explore conditional models, allowing one model to generate all types of supernovae
 - Industry generative models have come on a long way since GANs, explore new architectures e.g. transformers

Part 2: Fast Bayesian Inference

$$\mathcal{L} = \frac{1}{\sqrt{2\pi\sigma_f^2}} \exp \left[-\frac{(f_m - \hat{f})^2}{2\sigma_f^2} \right]$$

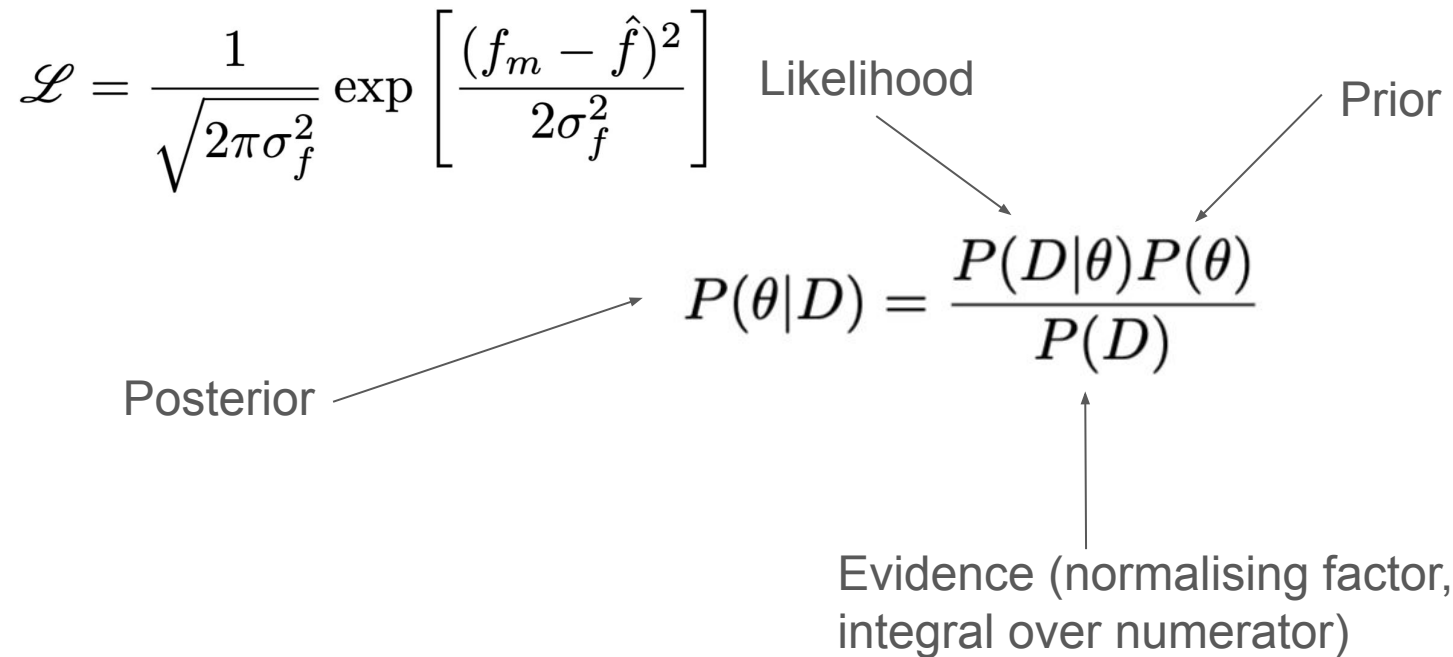
Likelihood

Prior

Posterior

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Evidence (normalising factor,
integral over numerator)



```
graph TD; Likelihood --> Posterior; Prior --> Posterior; Evidence --> Posterior;
```

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

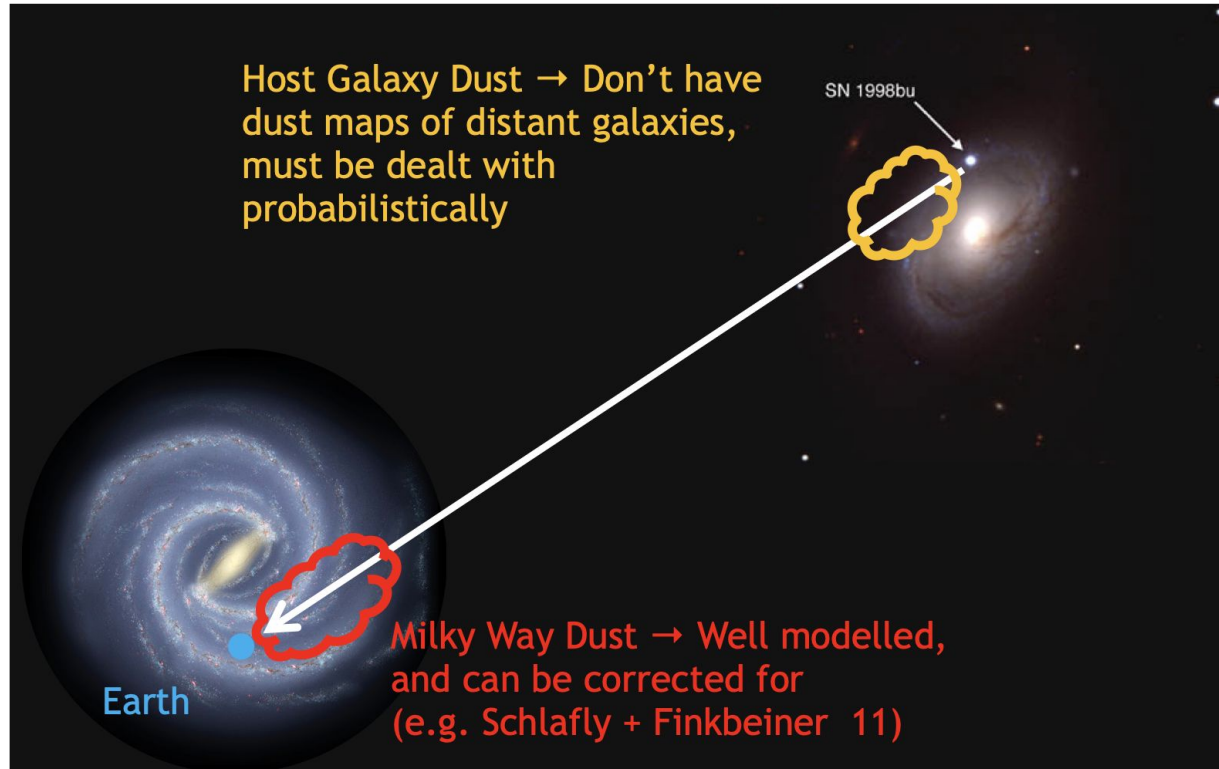
- For typical Bayesian inference, each supernova would be fit separately

$$R_V \sim U(1, 5)$$

- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

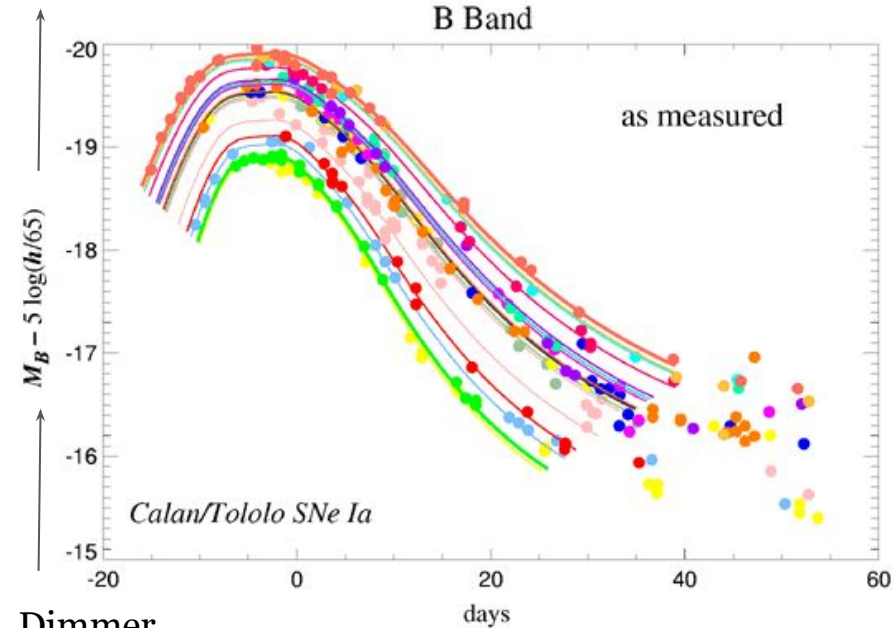
- We can model intrinsic variation and dust properties at the population level as two separate effects



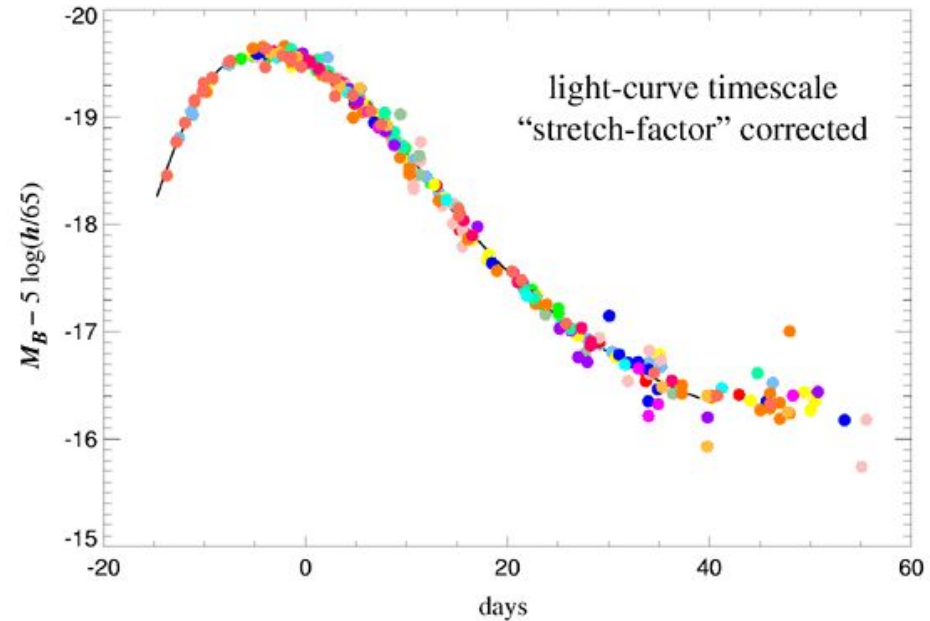
NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

SNe Ia - Standard(isable) Candles

Brighter



Dimmer



From Supernova Cosmology Project

Tripp formula

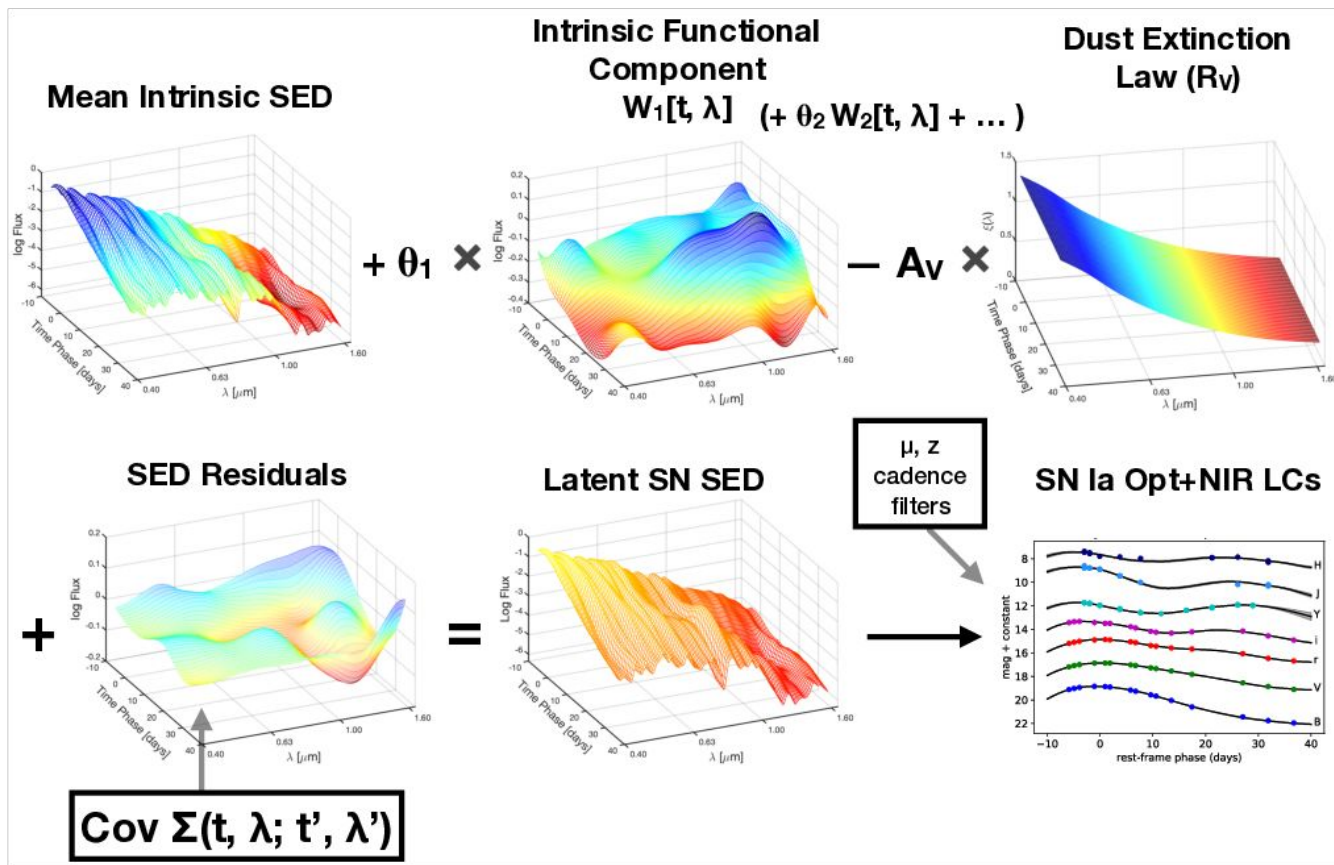
$$\mu_s = m_s - M + \alpha x_s + \beta c_s$$

Diagram illustrating the Tripp formula for standardizing supernova light curves:

- μ_s : Distance
- m_s : Observed brightness
- M : "Standard" brightness
- α : Stretch correction coefficient
- x_s : Stretch
- β : Colour correction coefficient
- c_s : Observed B-V colour

- “c” - one parameter for two independent effects (dust and intrinsic differences)
- Correctly handling dust (and SN Ia colour–luminosity correlations more generally) is key to correctly estimating SN Ia distances
- If intrinsic effect misattributed to dust, could lead to bias

The BayeSN Model

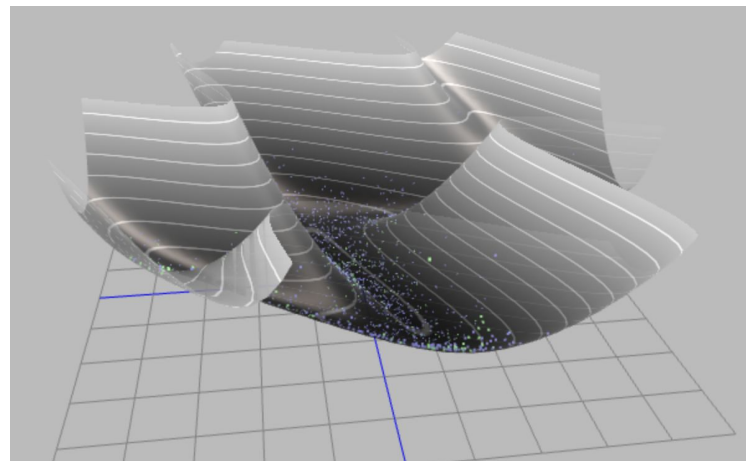


Mandel (2020)

- Lots of parameters (e.g. ~ 400 global, ~ 4200 latent parameters for Thorp 2021)
- Complex likelihood evaluation:
 - Compute SED model
 - Evaluate numerical integrals through different filters, for each time of observation and each supernova
- Computationally expensive

Isn't this a machine learning workshop?

- A model with thousands of parameters, fed through a series of large tensor operations to calculate a performance metric
 - Sounds a little like a neural network
- For a neural network, we use gradient-based optimisers to explore the parameter space, looking for the optimal solution
- For Bayesian inference, we usually want a posterior distribution over those parameters
 - Use Hamiltonian Monte Carlo (HMC) to sample posteriors or Variational Inference to fit them



Credit: Alex Rogozhnikov

Problem:

Scaling BayeSN for next generation data sets without compromising functionality

Solution:

pyro-ppl/**numpyro**



Probabilistic programming with NumPy powered by
JAX for autograd and JIT compilation to
GPU/TPU/CPU.

 82
Contributors

 529
Used by

 2k
Stars

 179
Forks



Jax

- Python package very similar to numpy, but includes JIT compilation at runtime for any device including GPUs
- Supports autodiff (automatic calculation of exact gradients), ideal for HMC or variational inference
- Very efficient matrix/tensor operations, perfect for numerical integrals

Numpyro

- Probabilistic programming package for Python built on Jax

Vectorized likelihood evaluation + GPUs = Fast Bayesian inference

Mandel+20:

- Trained on 79 *BVriYJH* light curves compiled in Avelino+19
- Previous training ~5 days → now ~20 minutes

Thorp+21:

- Trained on 157 *griz* light curves from Foundation DR1
- Previous training ~1 day → now ~10 minutes

1000 simulated optical SNe:

- Training (conditioning global parameters on data)
 - 45 mins
- Fitting (inference of supernova properties)
 - 15 mins
- Fitting with variational inference (work by Ana Sofia Uszoy)
 - 2.5 mins

- Many overlaps between neural networks and Bayesian inference
- Previously, applying hierarchical Bayesian models to data-sets $> 10,000$ SNe would have been computationally unfeasible
- The use of numpyro + GPUs makes this achievable in relatively short timescales
- We are able to scale Bayesian inference approaches to LSST-size data sets

Part 3: Simulation-based Inference

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

Likelihood

$$\mathcal{L} = \frac{1}{\sqrt{2\pi\sigma_f^2}} \exp \left[-\frac{(f_m - \hat{f})^2}{2\sigma_f^2} \right]$$

Gaussian likelihood - easy to calculate

$\mathcal{L} = ?$

Sometimes the likelihood is too complex to write down

Examples of selection criteria

- More than N observations with signal-to-noise $> 3/4/5$
- More than M observations before/after peak
- $|x_1| < 3$, $|c| < 0.3$
- Need a spectroscopic host redshift

We have developed tools to simulate supernova surveys well, but expressing their selection effects analytically is impossible

One example - ratio estimation

$$P(\Theta|\mathbf{D}) = \frac{P(\mathbf{D}|\Theta)P(\Theta)}{P(\mathbf{D})} \equiv r(\Theta, \mathbf{D})P(\Theta)$$

$$r(\Theta, \mathbf{D}) = \frac{P(\mathbf{D}|\Theta)}{P(\mathbf{D})} = \frac{P(\Theta, \mathbf{D})}{P(\Theta)P(\mathbf{D})}$$

Can estimate with
neural network

Just train a binary classifier on simulations -
“does this data set match this set of
parameters?”

- Many cases arise in astronomy where we can simulate an effect we cannot write down analytically
- SBI provides a tool for doing statistical inference in cases which would otherwise be impossible or require a number of (incorrect) assumptions
- Loads of SBI tools in existence:
 - Normalising Flows, Neural Posterior Estimation, Neural Ratio Estimation, ...
- SBI++ package (https://github.com/wangbingjie/sbi_pp) is a good place to get started if you are interested

Conclusions

- 1) Generative models have the potential to help combat data drift between real and simulated data, improving classifiers
- 2) Bayesian inference shares many similarities with neural networks which enable order-of-magnitude performance increases using GPUs, making it scalable for LSST-size data sets
- 3) Simulation-based inference will be an increasingly important, machine learning-based approach to statistical inference going forward

Postdoc opportunities at Cambridge in supernovae / astrostatistics /
data science... apply by January 2, 2024 (GMT)!

<https://www.jobs.cam.ac.uk/job/43803/>